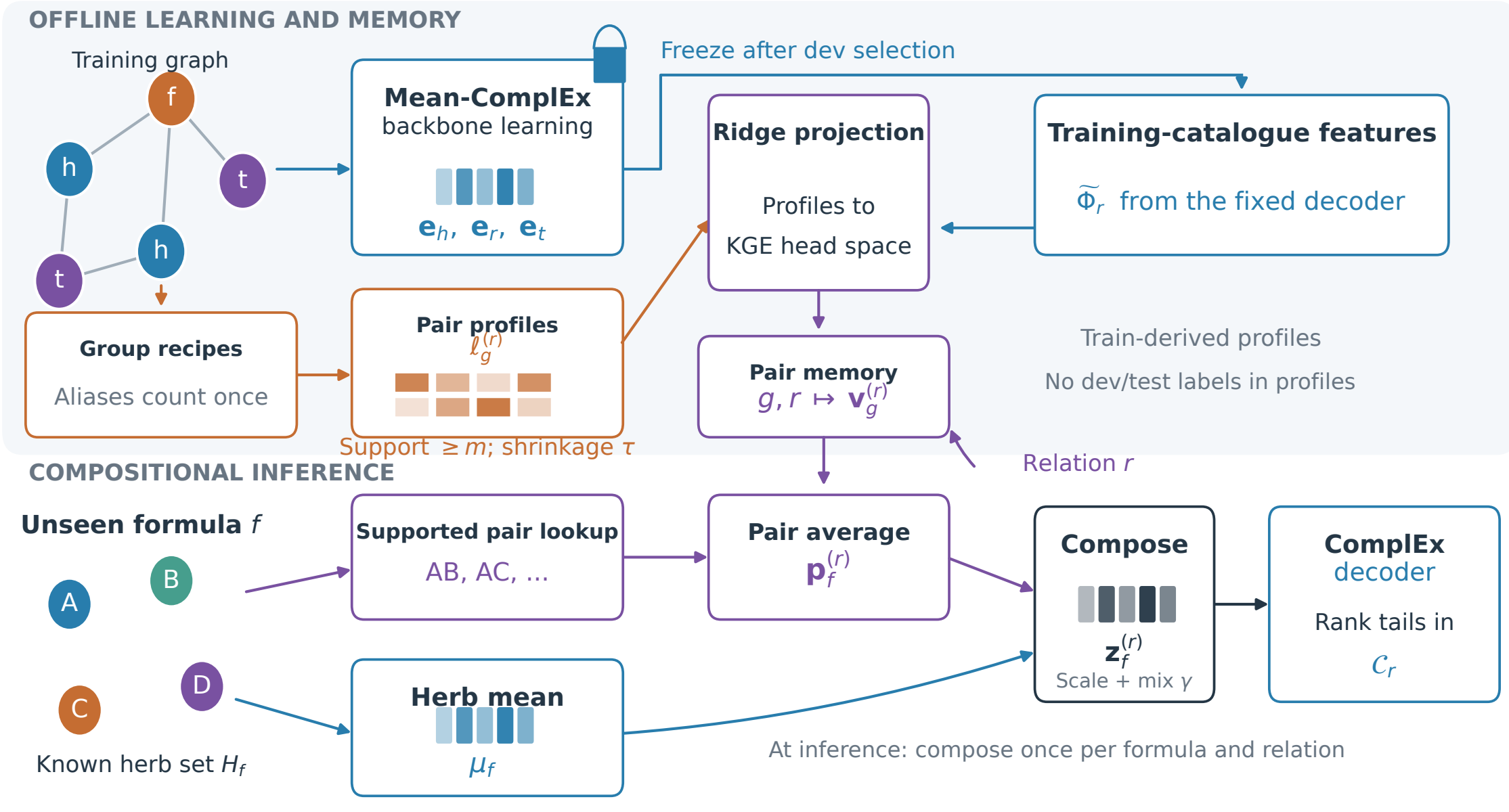
OFFLINE LEARNING AND MEMORY
Training graph
f
h
t
h
t
Mean-ComplEx
backbone learning
$\mathbf{e}_h$, $\mathbf{e}_r$, $\mathbf{e}_t$
Freeze after dev selection
Ridge projection
Profiles to
KGE head space
Training-catalogue features
$\widetilde{\Phi}_r$ from the fixed decoder
Group recipes
Aliases count once
Pair profiles
$\ell_g^{(r)}$
Support $\geq m$; shrinkage $\tau$
Pair memory
$g, r \mapsto \mathbf{v}_g^{(r)}$
Train-derived profiles
No dev/test labels in profiles
Relation $r$
COMPOSITIONAL INFERENCE
Unseen formula $f$
A
B
C
D
Known herb set $H_f$
Supported pair lookup
AB, AC, ...
Pair average
$\mathbf{p}_f^{(r)}$
Herb mean
$\mu_f$
Compose
$\mathbf{z}_f^{(r)}$
Scale + mix $\gamma$
ComplEx
decoder
Rank tails in
$\mathcal{C}_r$
At inference: compose once per formula and relation